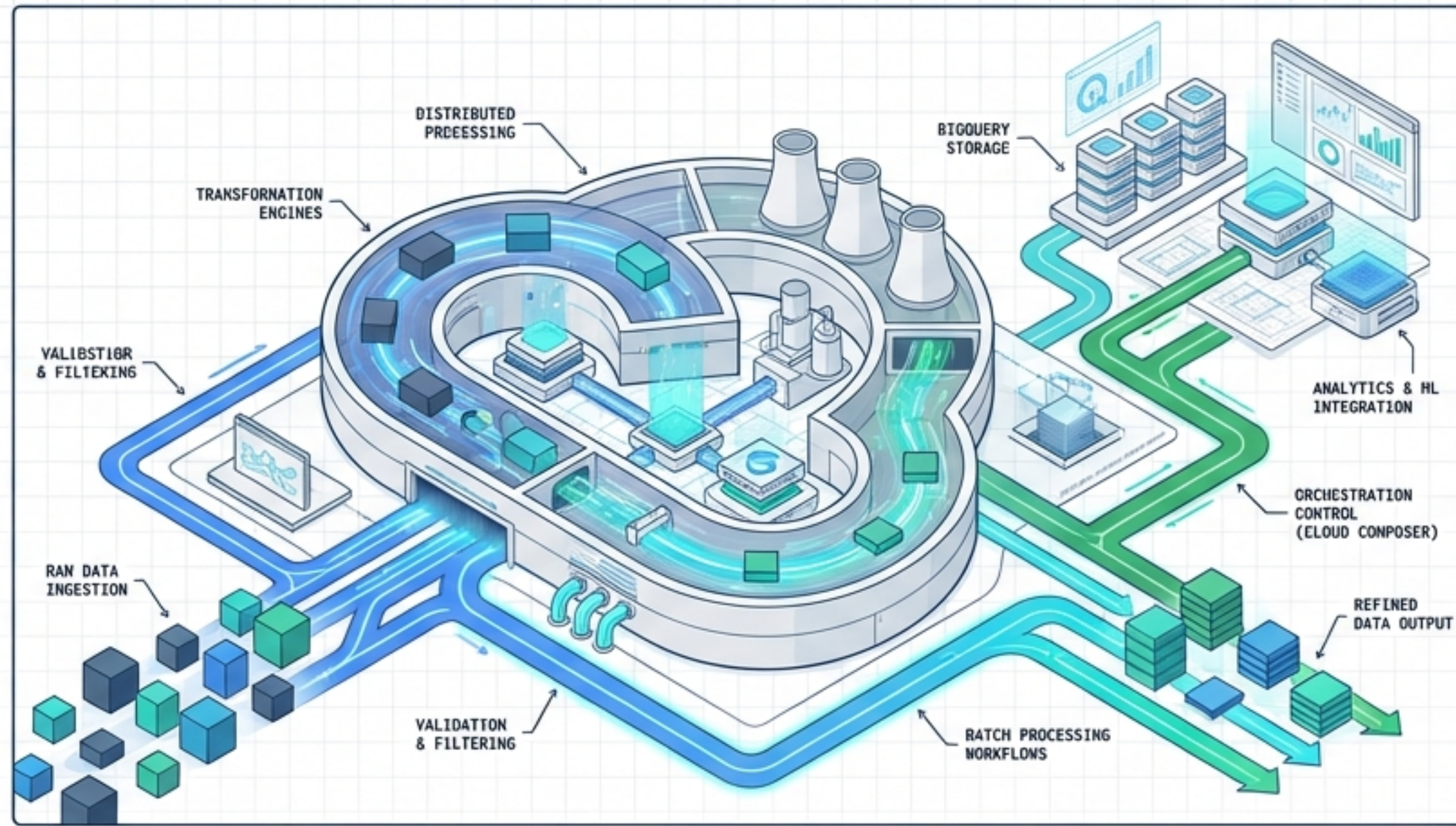




BUILD BATCH DATA PIPELINES ON GOOGLE CLOUD

A visual guide to designing, scaling, and orchestrating enterprise data pathways

The Data Engineer's Core Mandate

Build reliable pathways for data to move from raw sources to valuable insights.

The backbone of enterprise data strategy relies on processing large, finite datasets for critical operations like daily financial reconciliation.

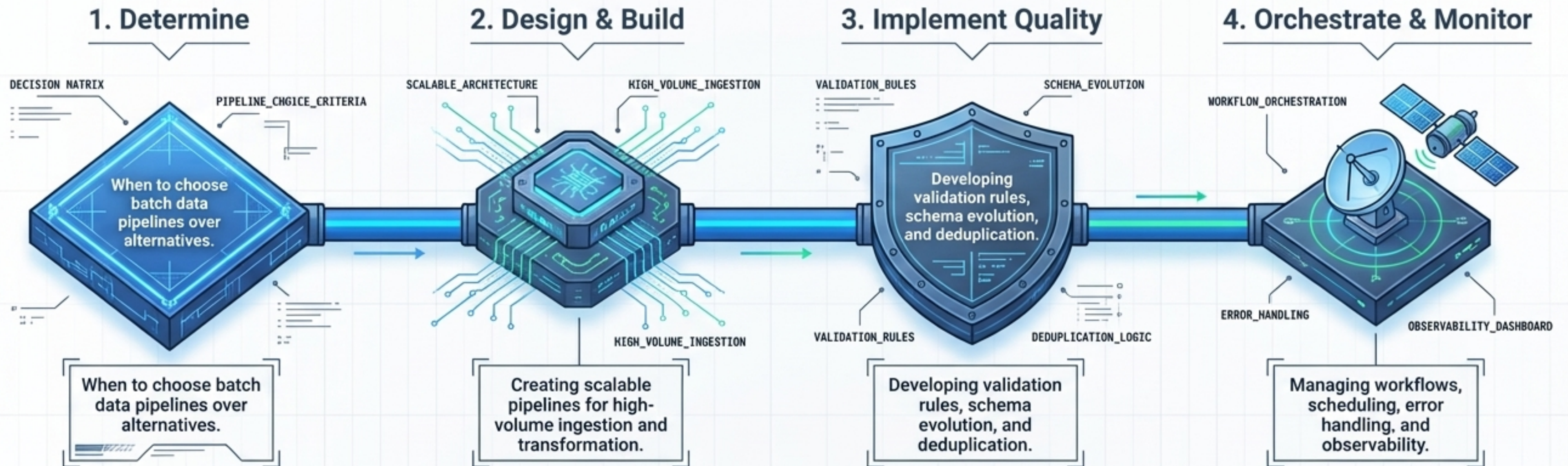
Cymbal Superstore



A fictional retailer processing millions of daily billing transactions. Their data arrives in scattered formats from diverse sources.

Their challenge: process massive scale accurately, reliably, and cost-effectively to enable timely invoicing and accurate financial reporting.

Course Syllabus Roadmap



The Anatomy of a Batch Pipeline

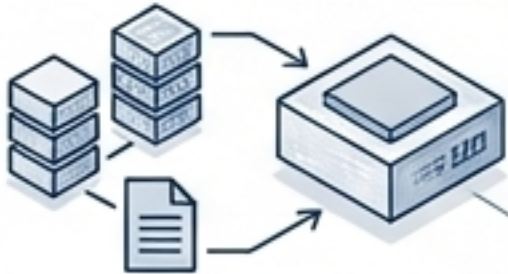
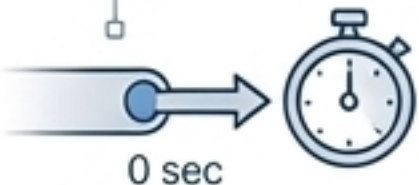
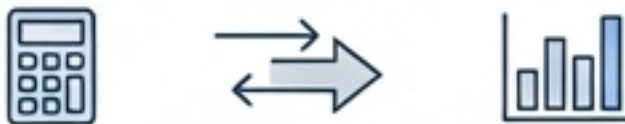


A sequence of processes designed to ingest, transform, and sink data in large, discrete chunks (batches) at scheduled intervals.

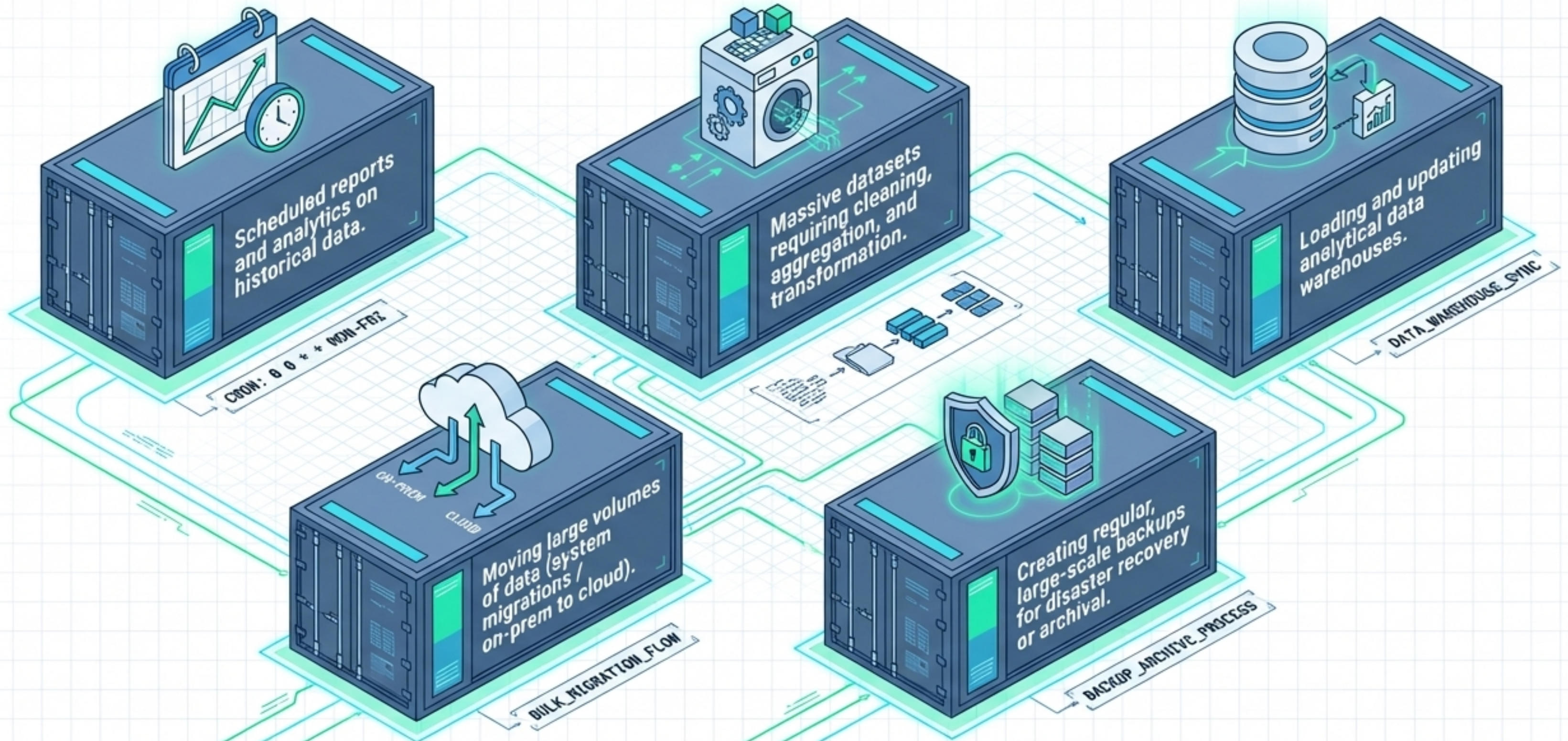


Optimized for massive throughput and efficiency over large volumes of historical or accumulated data.

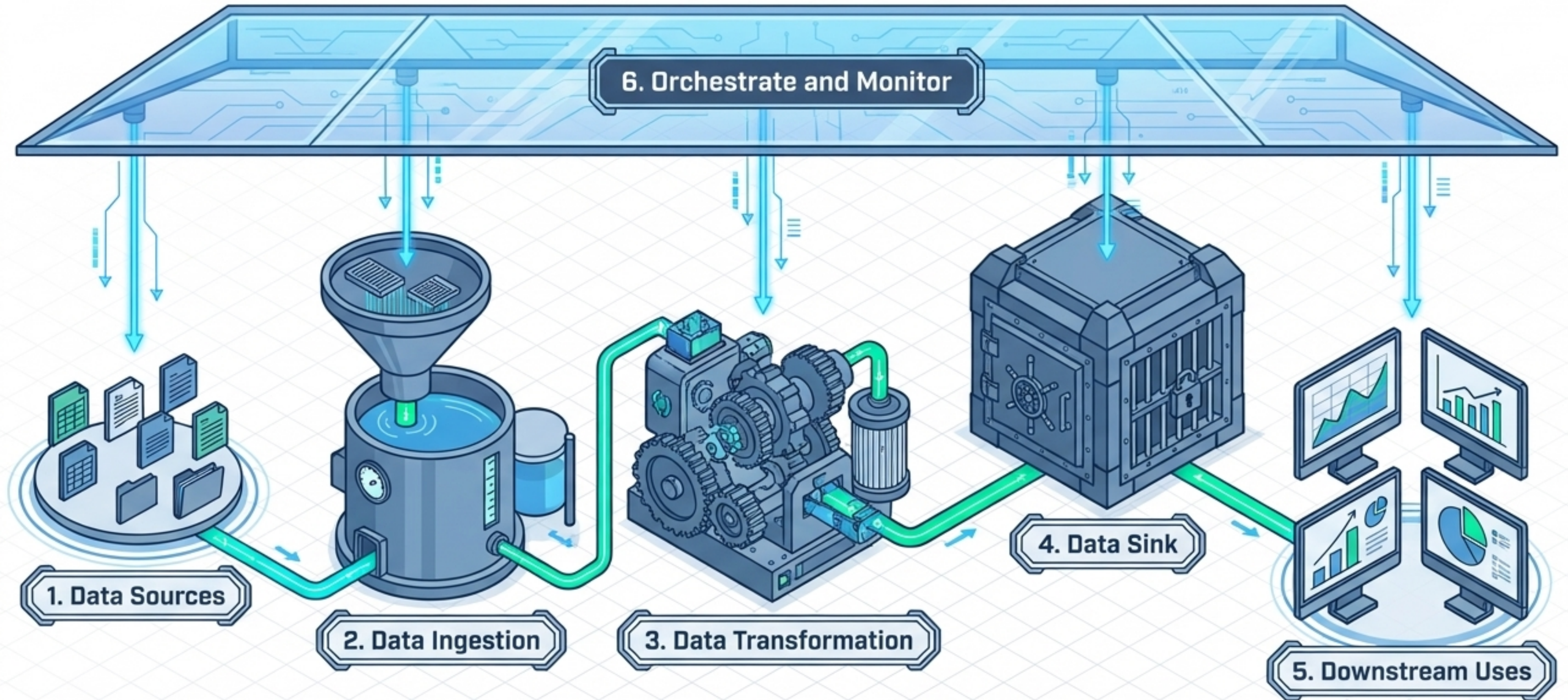
Diagnostic Matrix: Batch vs. Streaming

	Batch Processing	Real-Time Streaming
Data State	Large, finite datasets of historical or accumulated data. 	Unbounded, continuous data as it arrives. 
Processing Trigger	Runs at scheduled intervals (e.g., end-of-day). 	Runs continuously, reacting immediately.  Immediate response
Optimization Goal	High Throughput (processing massive volumes efficiently). 	Low Latency (delivering immediate results).  0 sec
Ideal Use Case	Financial reconciliation, massive data migrations, and scheduled analytics. 	Immediate live dashboards, fraud alerts, and anomaly detection. 

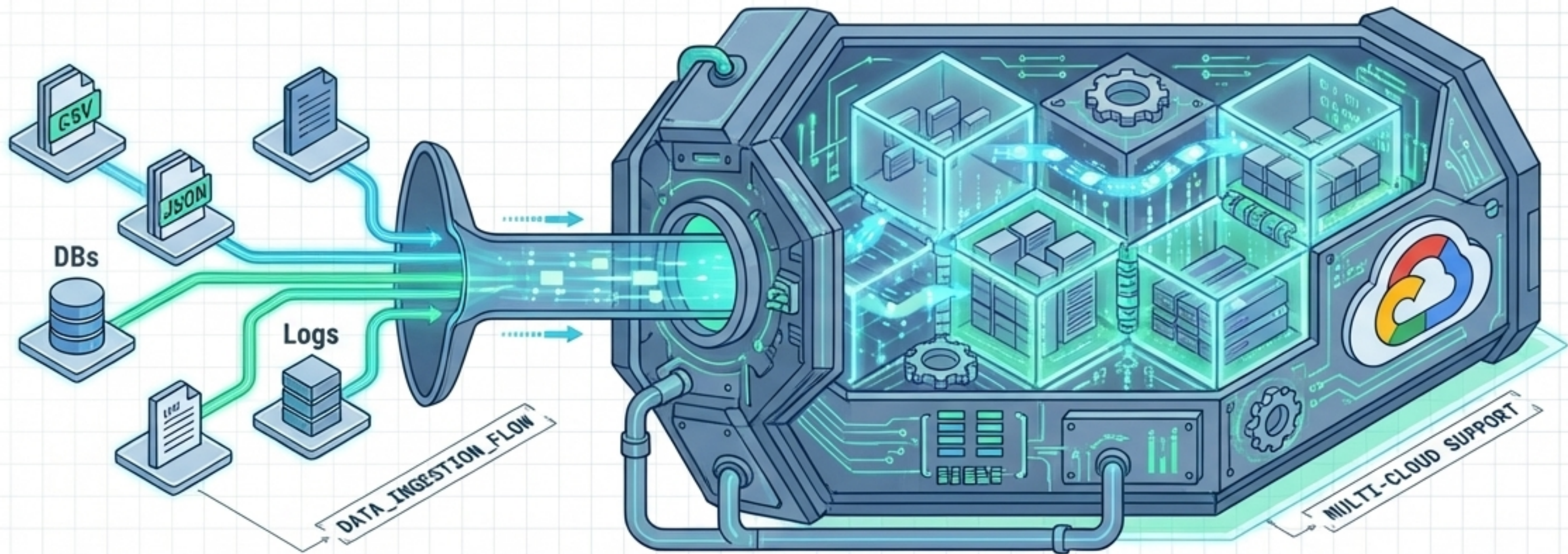
When to Choose Batch (Core Use Cases)



The Batch Pipeline Blueprint



Stage 1: Data Sources & Initial Ingestion



The Architecture

Acquiring raw data from origins into a central, temporary landing zone or staging area.

Note: Direct database ingestion is also an alternative to file drops.

Cymbal Superstore Implementation

Automating the collection of scattered daily sales transactions. CSVs and JSONs bypass costly duplicate copies via multi-cloud support, landing directly into Google Cloud Storage—the central hub for batch collection.

Programmatic Ingestion: The Automation Engine

Initializes the secure connection to Google Cloud.

```
from google.cloud import storage  
  
storage_client = storage.Client()  
bucket = storage_client.bucket(bucket_name)  
blob = bucket.blob(destination_blob_name)  
blob.upload_from_filename(source_file_name)
```

Locates the specific staging landing zone.

Executes the transfer of the discrete batch file into the cloud.

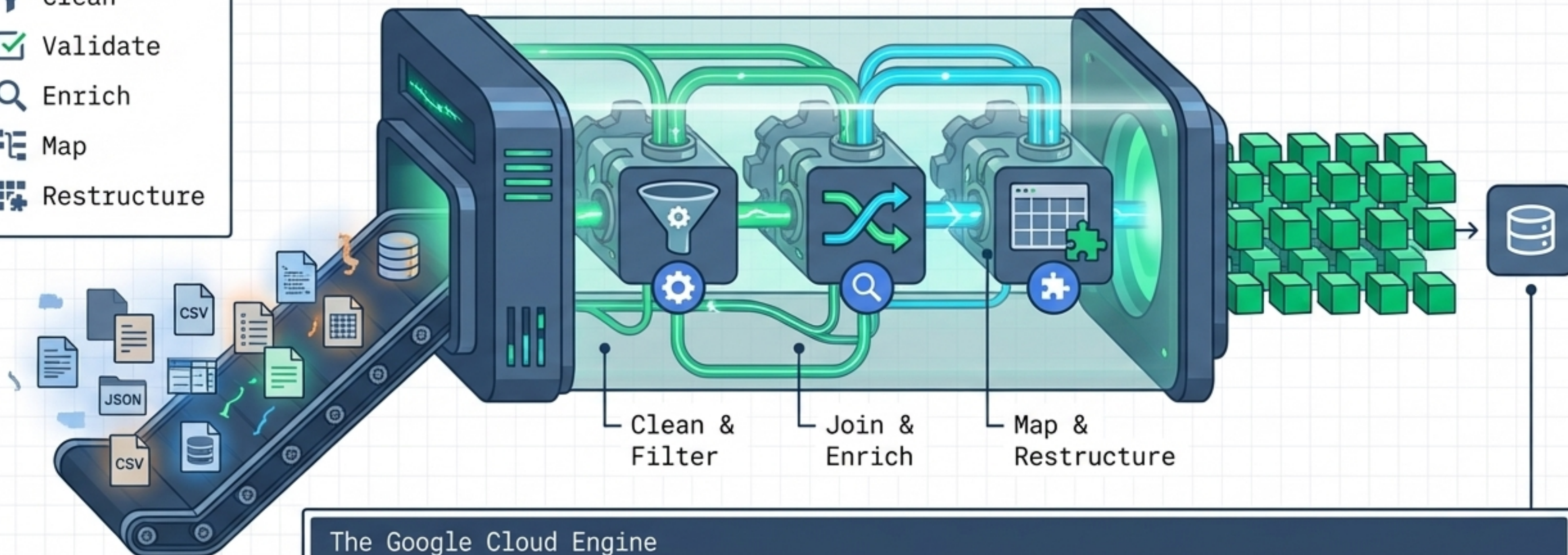


In enterprise environments, this ingestion is highly automated via libraries and scheduling tools, perfectly setting up the next stage of processing.

Stage 2: Data Transformation

Core Actions

-  Clean
-  Validate
-  Enrich
-  Map
-  Restructure

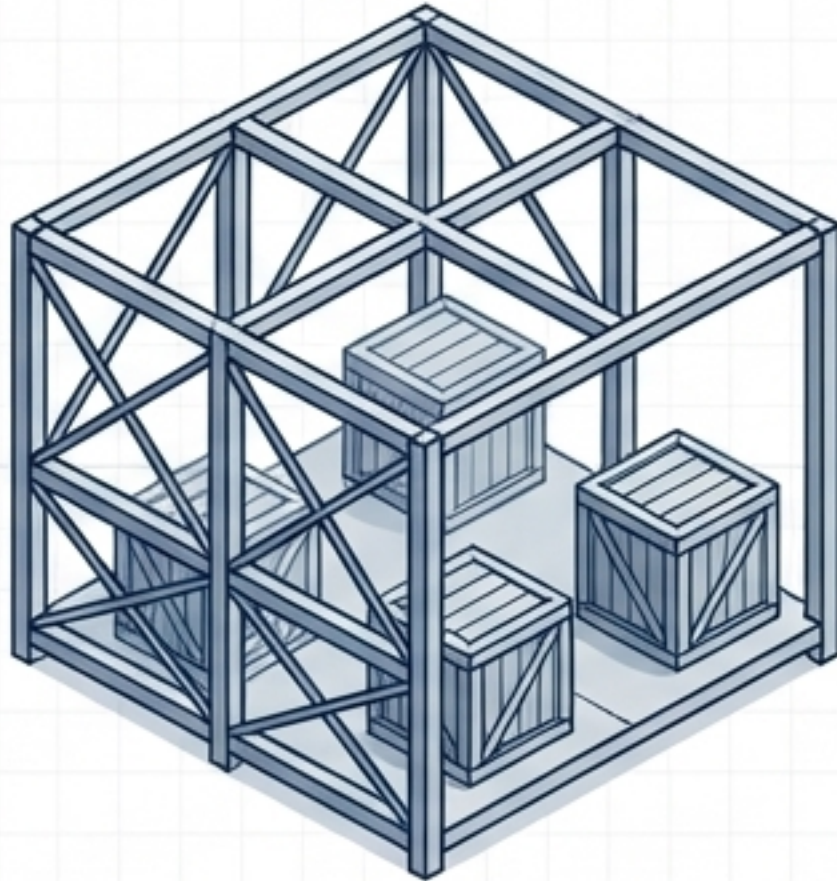


The Google Cloud Engine

Cymbal reads the raw data from Cloud Storage and processes it at massive scale using managed services: Dataflow for Apache Beam or Serverless for Apache Spark. These services handle all setup, scaling, and tear-down of job resources automatically.

Stage 3: The Data Sink

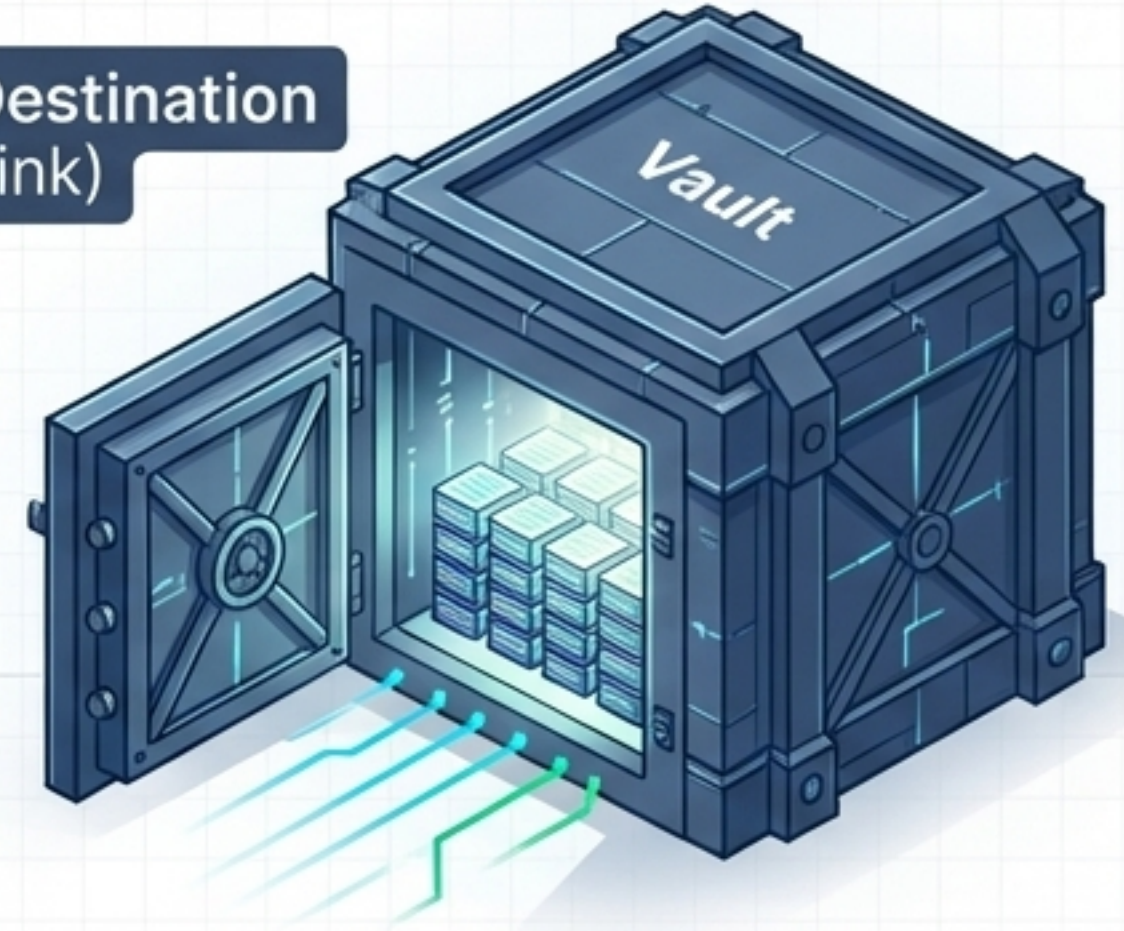
Temporary Loading Dock



Intermediate Storage

Holds data temporarily as it moves through the pipeline (e.g., the Cloud Storage landing zone).

Final Destination (The Sink)



Clean, structured data is loaded into storage optimized for specific uses.

-  **Data Warehouses:** 
BigQuery (Cymbal's choice, optimized for fast analytical queries on massive datasets).
-  **Data Lakes:** 
Cloud Storage (using formats like Apache Iceberg).

Stage 4: Downstream Value Extraction

How Cymbal Superstore consumes the processed batch data:



The Sink.
BigQuery Data Warehouse



Accurate Financial Reporting

Reconciliation and invoicing.



Interactive Business Dashboards

Historical sales analysis



Machine Learning Models

Predicting future sales trends.

The Control Room: Orchestrate & Monitor

Orchestration

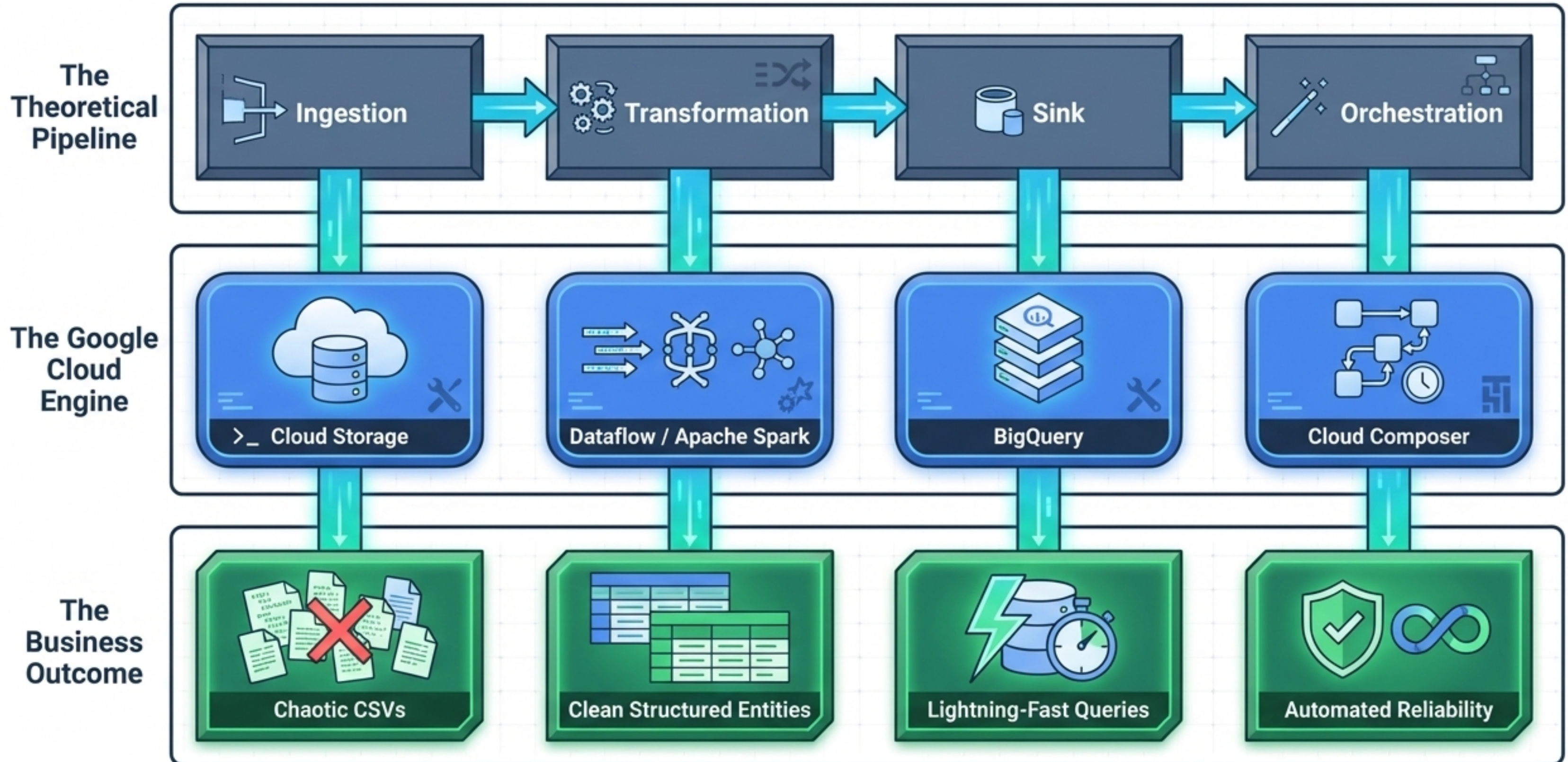
Managing and coordinating tasks. Ensuring jobs run in the correct order and handle dependencies automatically using tools like Cloud Composer.

Monitoring

Tracking system health, performance, and data quality. Alerting on errors and ensuring strict data integrity for reliable critical operations.



The Complete Cymbal Superstore Stack



Diagnostic Check



A company needs to generate complex financial reports and update its data warehouse. The process must run daily after business hours, processing all of the day's transactions at once. The job is optimized for throughput, and immediate results are not required.



Diagnosis: Batch Processing

This scenario requires processing a large, finite dataset at a scheduled interval. It correctly prioritizes high throughput (volume) over low latency (speed), making batch architecture the perfect fit.